

Q1 - 26 June - Shift 2

If $\int \frac{1}{x} \sqrt{\frac{1-x}{1+x}} dx = g(x) + c, g(1) = 0$, then $g\left(\frac{1}{2}\right)$ is equal

Space for your notes:

to :

(A) $\log_e \left(\frac{\sqrt{3}-1}{\sqrt{3}+1} \right) + \frac{\pi}{3}$ (B) $\log_e \left(\frac{\sqrt{3}+1}{\sqrt{3}-1} \right) + \frac{\pi}{3}$

(C) $\log_e \left(\frac{\sqrt{3}+1}{\sqrt{3}-1} \right) - \frac{\pi}{3}$ (D) $\frac{1}{2} \log_e \left(\frac{\sqrt{3}-1}{\sqrt{3}+1} \right) - \frac{\pi}{6}$

Q2 - 27 June - Shift 1

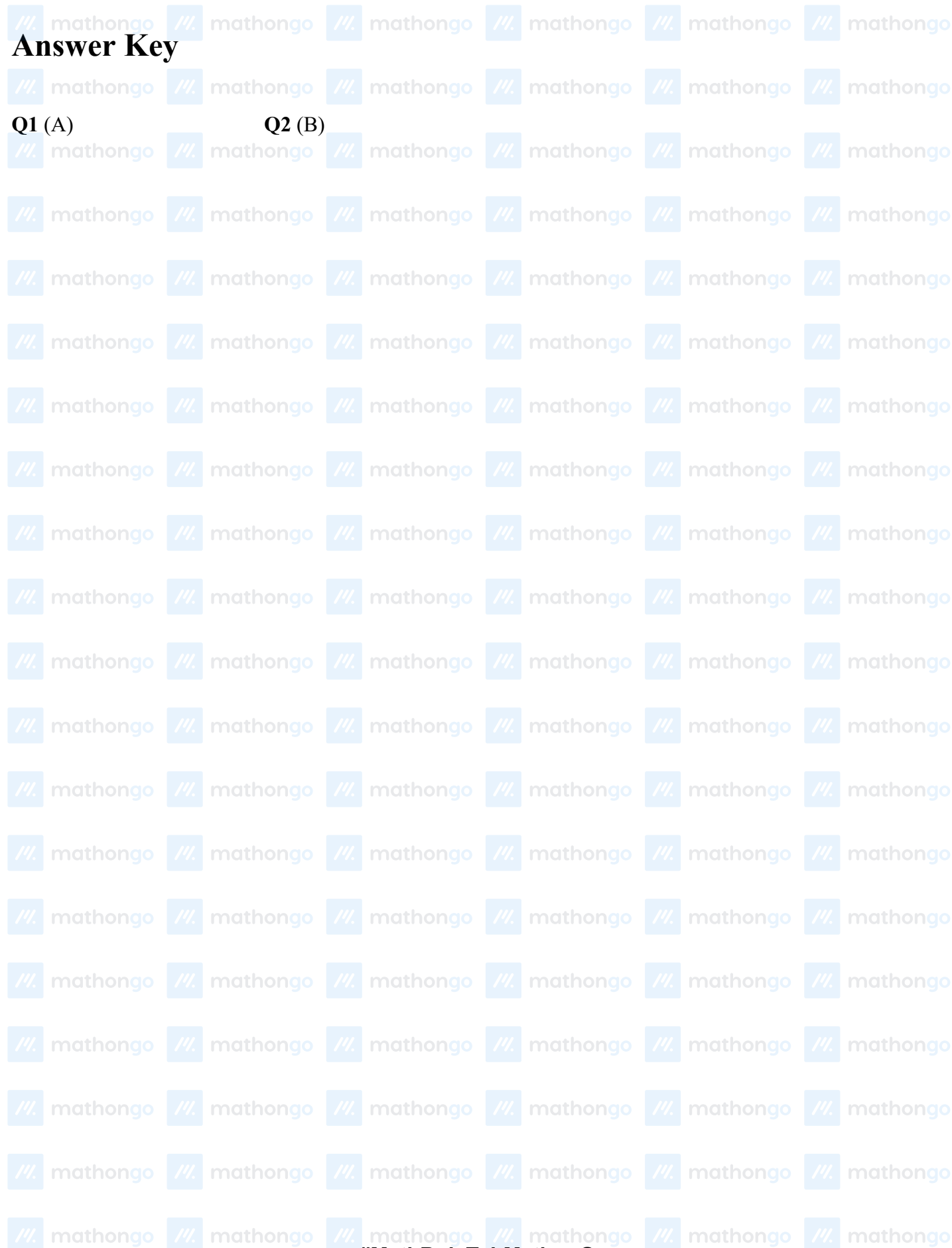
$\int \frac{(x^2+1)e^x}{(x+1)^2} dx = f(x)e^x + C$, Where C is

Space for your notes:

constant, then $\frac{d^3 f}{dx^3}$ at $x = 1$ is equal to :

(A) $-\frac{3}{4}$ (B) $\frac{3}{4}$

(C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

Answer Key**Q1 (A)****Q2 (B)****#MathBoleTohMathonGo**

Indefinite Integration

JEE Main 2022 (June) Chapter-wise Qs Bank

Hints and Solutions

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$$\int \frac{1}{x} \sqrt{\frac{1-x}{1+x}} dx = g(x) + c$$

Put $x = \cos 2\theta$

$$dx = -2\sin 2\theta \cdot d\theta$$

$$= \int \frac{1}{\cos 2\theta} \tan \theta (-4 \sin \theta \cdot \cos \theta) d\theta$$

$$= \int \frac{1}{\cos 2\theta} (-4 \sin^2 \theta) d\theta$$

$$= -2 \int \frac{1 - \cos 2\theta}{\cos 2\theta} d\theta$$

$$= -\frac{2}{2} \ln |\sec 2\theta + \tan 2\theta| + 2\theta + c$$

$$= \ln |\sec 2\theta - \tan 2\theta| + 2\theta + c$$

$$= \ln \left| \frac{1 - \sin 2\theta}{\cos 2\theta} \right| + \cos^{-1} x + c$$

$$= \ln \left| \frac{1 - \sqrt{1-x^2}}{x} \right| + \cos^{-1} x + c$$

$$\therefore g(1) = 0$$

$$g(x) = \ln \left| \frac{1 - \sqrt{1-x^2}}{x} \right| + \cos^{-1} x$$

$$g\left(\frac{1}{2}\right) = \ln |2 - \sqrt{3}| + \frac{\pi}{3}$$

$$g\left(\frac{1}{2}\right) = \ln \left| \frac{\sqrt{3}-1}{\sqrt{3}+1} \right| + \frac{\pi}{3}$$

 Q2 (B)					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

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$$\int \left(\frac{x^2 + 1}{(x+1)^2} \right) e^x \cdot dx$$

$$= \int \left(\frac{x^2 - 1 + 2}{(x+1)^2} \right) e^x dx$$

$$= \int \left(\frac{x-1}{x+1} + \frac{2}{(x+1)^2} \right) e^x dx$$

$$= \int (f(x) + f'(x)) e^x dx$$

$$= f(x) e^x + c$$

$$\text{Where } f(x) = \frac{x-1}{x+1}$$

$$f'(x) = \frac{2}{(x+1)^2}$$

$$f''(x) = \frac{-4}{(x+1)^3}$$

$$= \frac{12}{(x+1)^4}$$

$$f''(1) = \frac{12}{16}$$

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